

Homework set 2

Action principle for a relativistic particle

Consider a free particle moving along $x^\mu(t)$ in Minkowski spacetime

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$$

Take as the action

$$S = -m \int ds = -m \int dt \left(\frac{ds}{dt} \right)$$

vary it and find the equations of motion.

Coulomb gauge:

Consider classical Maxwell fields.

- Expand the Lagrangian density in the presence of a current in terms of

$A^\mu = (\phi, \vec{A})$ with ϕ the electrostatic potential & $\vec{A}$ satisfying

the Coulomb gauge $\vec{\nabla} \cdot \vec{A}(x) = 0$.

- Vary the action with respect to φ and $\vec{A}$ to find their equations of motion
- Find the solution to the equations of motion using Fourier transform.

Spontaneous Symmetry Breaking:

Consider a complex relativistic scalar field with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \Phi)^\dagger (\partial^\mu \Phi) + \lambda(\Phi^\dagger \Phi - \phi_0)^2$$

- Find the classical equations of motion
- Consider constant field configurations and find the minima of action.
- Expand the field $\Phi(x^\mu)$ around this minimum to the second order and find the mass of the excitations.
- Is there a massless mode?